

Home Work 4
II B. Tech. (Common)
Mathematics-II (NMCI102)

Notations:

$M_{m \times n}(\mathbb{F})$: The set of all matrices of order $m \times n$ with entries from $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

$M_n(\mathbb{F})$: The set of all matrices of order $n \times n$ with entries from $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

$P(\mathbb{R})$ or $\mathbb{R}[x]$: The set of all polynomials in one variable with real coefficients.

$P_n(\mathbb{R})$ or $\mathbb{R}_n[x]$: The set of all polynomials of degree at most n in one variable with real coefficients.

1. Find a basis and the dimension of the following vector spaces.
(a) $\mathbb{R}^n$ over $\mathbb{R}$ (b) $\mathbb{C}$ over $\mathbb{R}$ (c) $P(\mathbb{R})$ over $\mathbb{R}$ (d) $M_{m \times n}(\mathbb{R})$ over $\mathbb{R}$
2. Find a basis and the dimension of the following subspaces W of the corresponding vector spaces.
(a) $W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$ (b) $W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0, 2x + y + z = 0\}$
(c) $W = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2 \times 2}(\mathbb{R}) : b = c = 0 \right\}$ (d) $W = \{A \in M_{2 \times 2}(\mathbb{R}) : A = A^T\}$
(e) $W = \{A \in M_{2 \times 2}(\mathbb{R}) : A = -A^T\}$ (f) $W = \{A \in M_{n \times n}(\mathbb{R}) : \text{trace}(A) = 0\}$
(g) Fix $a \in \mathbb{R}$, $W = \{f(x) \in P_n(\mathbb{R}) : f(a) = 0\}$ (h) $W = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2 \times 2}(\mathbb{R}) : a + b = 0 \right\}$.
(i) $W = \{A = (a_{ij}) \in M_n(\mathbb{R}) \mid a_{ij} = 0 \text{ if } i \text{ is even}\}$ (j) $W = \{A \in M_{2 \times 2}(\mathbb{C}) : A = \bar{A}^T\}$ over $\mathbb{R}$.
3. Which of the following is a basis of $\mathbb{R}^3$ over $\mathbb{R}$?
(a) $\{(1, 1, 0), (5, 5, 1), (8, 8, 4)\}$ (b) $\{(1, 1, 0), (0, 1, 1), (1, 0, 1)\}$ (c) $\{(1, 1, 0), (0, 1, 1), (1, 0, 1), (0, 0, 1)\}$
(d) $\{(1, 1, 0), (0, 0, 3)\}$.
4. For what real values of k , does the the set $\{(k, 0, 1), (1, k + 1, 1), (1, 1, 1)\}$ forms a basis of $\mathbb{R}^3$ over $\mathbb{R}$?
[Ans. $k \neq 0, 1$]
5. Give three different basis for $M_{2 \times 2}(\mathbb{R})$ and $\mathbb{R}^2$ over $\mathbb{R}$.
6. Find all subspaces of $\mathbb{R}^2$ and $\mathbb{R}^3$.
7. Using the row echelon form of a matrix, find the row rank, the column rank, the rank and the nullity of the following matrices. Also, find bases for the row space, the column space, and the null space of the matrices.
(a) $A = \begin{bmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 6 & 9 \\ 1 & 1 & 2 & 6 \end{bmatrix}$ (b) $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 1 & 0 \end{bmatrix}$ (c) $A = \begin{bmatrix} -2 & 0 & 0 & 3 \\ 1 & 5 & 3 & 0 \\ 3 & 2 & 1 & 6 \\ 3 & 5 & 3 & -3 \end{bmatrix}$ Answers: (a) 2 (b) 2 (c) 3
8. Let V be a vector space over the field $F = \mathbb{R}$ or $\mathbb{C}$. Suppose $\{v_1, \dots, v_k\}$ is linearly independent in V and $w \in V$. Show that $\{v_1, \dots, v_k, w\}$ is linearly independent if and only if $w \notin \text{Span}(\{v_1, \dots, v_k\})$.
9. Extend the set $\{(1, 2, 1, 2), (2, 1, 2, 1)\}$ to a basis of $\mathbb{R}^4$ over $\mathbb{R}$.
10. Find the dimension of the solution space of the following set of homogeneous system of linear equations:
 $3x + 4y - z - 6w = 0, 2x + 3y + 2z - 3w = 0, 2x + y - 14z - 9w = 0, x + 3y + 13z + 3w = 0.$